BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2019

CHEMISTRY

Past Paper 2019 • Paper Code: A

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – C (MULTIPLE CHOICE QUESTIONS)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** A

Note: Q. 16: Choose the correct answer from the given options and write the same in your answer copy.

Q1. Solution having higher concentration of OH^- than that of H_3O^+ shows pH.

(a) Less than 7
(b) Greater than 7
(c) Equal to 7
(d) Zero

Q2. pH of H_2O changes either by absorbing silicates of glass or atmospheric

(a) CO_2
(b) O_2
(c) CO
(d) N_2

Q3. By looking at the following typical acid dissociation constant values, which acid is the strongest among them? ACID | Ka (mol/dm³): A = 4.8×10^{-10}; B = 1.8×10^{-4}; C = 1.8×10^{-5}; D = 1.8×10^{-3}

(a) Acid A
(b) Acid B
(c) Acid C
(d) Acid D

Q4. Rate of reaction dx/dt is represented by units

(a) $mole \cdot dm^3 \cdot s^{-1}$
(b) $mole^{-1} \cdot dm^3 \cdot s^{-1}$
(c) $mole \cdot dm^{-3} \cdot s^{-1}$
(d) $mole^{-1} \cdot dm^{-3} \cdot s^{1}$

Q5. The simplified result of 15.88 ÷ 7.95 by considering the rules of significant digits is:

(a) 1.997
(b) 2.00
(c) 1.99
(d) 1.9

Q6. The decomposition of glycerin at its boiling point 290°C at 760 torr is prevented by heating

(a) Below 760 torr
(b) above 760 torr
(c) at 1 atm
(d) at 14.7psi

Q7. $NaNO_3$ is isomorphous of:

(a) $ZnSO_4$
(b) $NiSO_4$
(c) $BaSO_4$
(d) $CaCO_3$

Q8. The bond distances of C − C and C = C one 1.54 Å and 1.34 Å respectively, expected bond distance of C ≡ C is:

(a) 1.55 Å
(b) 1.56 Å
(c) 1.20 Å
(d) 1.57 Å

Q9. In ammonia, nitrogen contains one lone pair and in water, oxygen contains two lone pairs, hence bond angle of water molecule is:

(a) Greater than ammonia
(b) Less than ammonia
(c) Same as ammonia
(d) None of these

Q10. What would be effect of increasing pressure at equilibrium on following reaction? $N_2 + 3H_2 \rightleftharpoons 2NH_3$

(a) Shift reaction Forward
(b) Shift reaction backward
(c) Have no effect
(d) Stop the reaction

Q11. Which of the following parameters is lowered down by the catalyst in attaining the chemical equilibrium?

(a) ΔE
(b) ΔH
(c) [illegible in source scan]
(d) E.A

Q12. When HCl gas is passed through NaCl saturated solution, the solubility of NaCl:

(a) Does not change
(b) It decomposes
(c) Decreases
(d) Increases

Q13. Radii of Cl^-, O^{2-}, Ca^{2+} and Al^{3+} are 1.81 Å, 1.40 Å, 0.99 Å and 0.50 Å respectively, which of these will be readily hydrated?

(a) O^{2-}
(b) Al^{3+}
(c) Cl^-
(d) Ca^{2+}

Q14. Which of the following pair form buffer solution.

(a) $NH_4OH + NH_4Cl$
(b) NaOH + NaCl
(c) $NaOH + CH_3COONa$
(d) $NaCl + CH_3COOH$

Q15. The series of lines observed in the infra-red region with reference to atomic spectrum of hydrogen is:

(a) Lyman series
(b) Balmer series
(c) Pfund series
(d) Paschen series

Q16. While writing electronic configuration of nitrogen as following; which rule is violated? N = (Z = 7) $1s^2$, $2s^2$, $2px^2$, $2py^1$, $2pz$

(a) Pauli's exclusion principle
(b) Aufbau principle
(c) Hund's Rule
(d) None of these

Q17. Which of the following is physical method used for the determination of rate of chemical reaction?

(a) Concentration
(b) Calorimetric method
(c) Titration
(d) Weight

SECTION – A (Short Answer Questions)

Time Allowed: 2 Hours and 40 Minutes | **Maximum Marks:** 40

Note: ATTEMPT ANY EIGHT (08) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q1. Explain the significances of Pauli's exclusion principle in relation to electronic configurations of atoms

Q2. Derive the value of Bohr's atomic radius and calculate radius of 2nd orbit of hydrogen atom.

Q3. How hydrogen is used as reference electrode in determining electrode potential.

Q4. 250 cm^3 of Hydrogen when cooled from 127°C to −27°C by keeping pressure unchanged, what would be the new volume of gas?

Q5. 500 J of work is done on a system which gives off 300 J of heat. Find value of ΔE for the process, also define state and state function.

Q6. How following factors affect on the rate of reaction? (i) Surface area of reactants (ii) Concentration of reactants (iii) Temperature

Q7. Write short note on Surface tension.

Q8. Explain with chemical method that how rate of chemical reaction is determined.

Q9. Define Oxidation and Reduction also balance the following equations by ion-electron method. $Cr_2O_7^{-2} + I^{-1} \rightarrow Cr^{+3} + IO_3^-$ (acidic)

Q10. Define unit cell and give the names of crystal system and also write axis and angles of different crystal system.

Q11. Derive an expression of the Equilibrium Constant for $2p + 3q \rightleftharpoons 2R$

Q12. Calculate the heat of formation of acetylene $2C + H_2 \rightarrow C_2H_2$ from following data: (i) $C + O_2 \rightarrow CO_2$ $\Delta H = -393.7$ KJ/mol (ii) $H_2 + 1/2\ O_2 \rightarrow H_2O$ $\Delta H = -284.3$ KJ/mol (iii) $C_2H_2 + 5/2\ O_2 \rightarrow 2CO_2 + H_2O$ $\Delta H = -1296$ KJ/mol

SECTION – B (Long Answer Questions)

Time Allowed: 2 Hours and 40 Minutes	**Maximum Marks:** 28

> Note: ATTEMPT ANY TWO (02) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q13.

(a) [07] Commencing with KE = 1/2 mv^2, PE = $-Ze^2/r$, derive an expression for energy of electron in nth orbit, frequency and wave number?

(b) [07] Distinguish between: (i) Reversible and Irreversible reaction (ii) Atomic Orbital and Molecular orbital

Q14.

(a) [07] Define concept of hybrid orbital. Discuss shapes of tetrahedral molecules in terms of electron pair repulsion model.

(b) [07] A mixture of 40g each of He and Ne has total pressure of 500 torr. Calculate the individual pressure of both gases.

Q15.

(a) [08] Define Buffer solution. Give its preparation and buffer action with its applications.

(b) [06] With the help of following data, determine the empirical formula of given compound. Element | % | Mole Ratio | Simplest Ratio: C 52.108; H 13.77; O 34.77